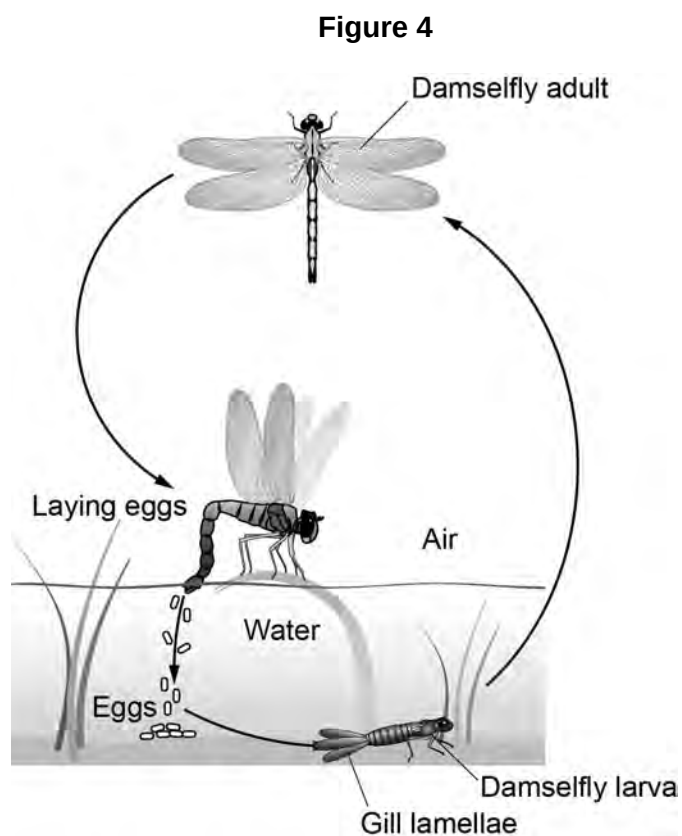


Figure 4 shows the stages of development of an insect called a damselfly.



09 . 1

The adult damselfly uses a tracheal system for gas exchange.

Explain **three** ways in which an insect's tracheal system is adapted for efficient gas exchange.

[3 marks]

1 _____

2 _____

3 _____

09 . 2

The damselfly larva is a carnivore that actively hunts prey. It has gills to obtain oxygen from water.

Some other species of insect have larvae that are a similar size and shape to damselfly larvae and also live in water. These larvae do **not** actively hunt prey and do **not** have gills.

Explain how the presence of gills adapts the damselfly to its way of life.

[2 marks]

09 . 3

A scientist measured the size of each gill lamella of the gills of 40 damselfly larvae. His results are shown in **Table 5**.

Table 5

Mean width / mm ($\pm$ uncertainty / mm)	1.61 ($\pm$ 0.19)
Mean length / mm ($\pm$ uncertainty / mm)	6.12 ($\pm$ 0.41)

Calculate the mean surface area of **one side** of one gill lamella. Assume that a gill lamella is rectangular and give your answer to an **appropriate number of significant figures**.

Include the percentage error (uncertainty) of surface area in your answer. Show your working.

[3 marks]

Mean surface area = _____

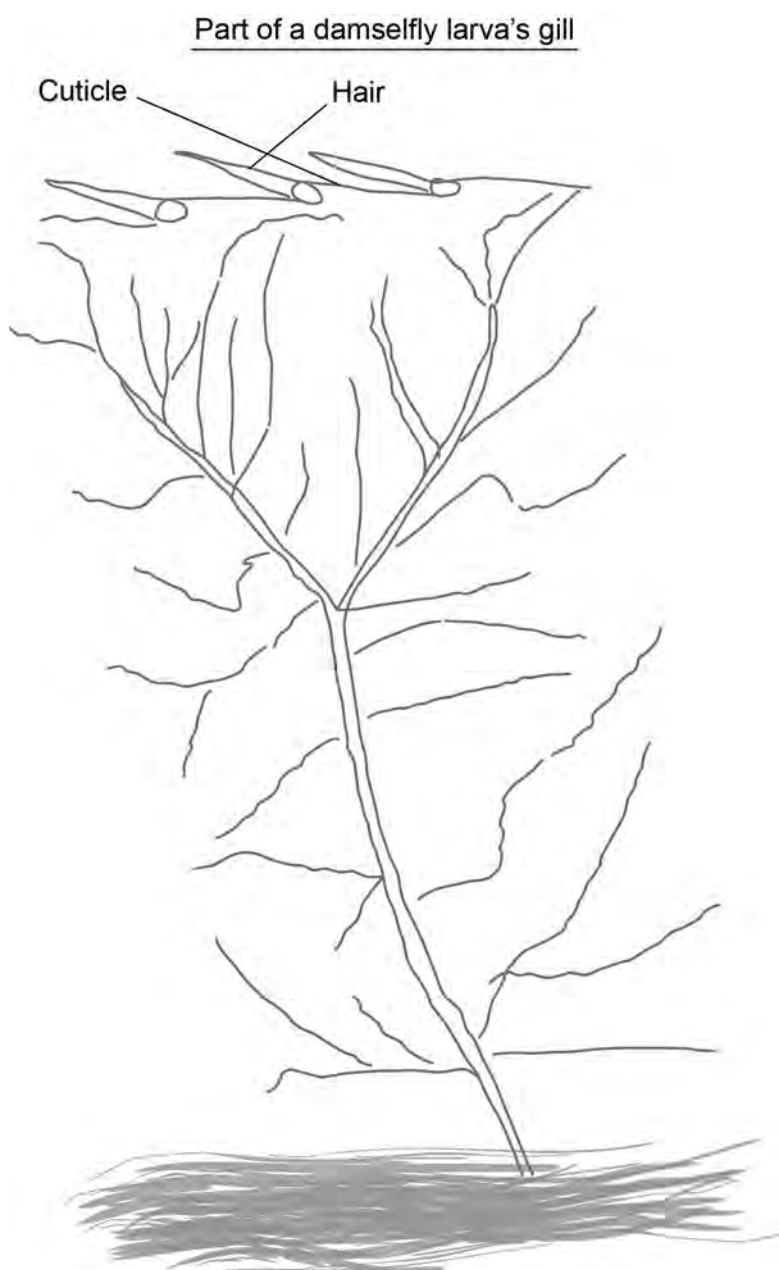
Percentage error (uncertainty) of surface area = _____

09 . 4

A student used an optical microscope to observe part of a damselfly larva gill.

Figure 5 shows the drawing the student produced.

Figure 5



Suggest **two** ways the student could improve the quality of her scientific drawing of this gill.

[2 marks]

1 _____

2 _____

1

Turn over for the next question

Turn over ►

